Deployment Plan - Hospital Management System

1. Introduction

1.1 Purpose

This document outlines the deployment strategy for the Hospital Management System (HMS). It ensures a seamless transition from development to production while maintaining high standards of reliability, security, and scalability.

1.2 Scope

The deployment plan covers the release process, environments, CI/CD pipelines, rollback strategies, monitoring, and security measures specific to the HMS. The system encompasses key modules such as Doctor, Patient, Prescription, Appointment, and Polyclinic.

1.3 Target Audience

- DevOps Engineers
- System Administrators
- Developers
- QA Engineers

2. Deployment Architecture

2.1 Infrastructure Overview

- **Cloud Provider:** AWS / Google Cloud / Azure
- **Compute Services:** EC2 Instances / Kubernetes / Docker Containers
- **Database:** PostgreSQL / MySQL (Managed Service)
- **Storage:** S3 for static files, EFS for persistent data
- **Caching:** Redis / Memcached
- **Load Balancing:** Elastic Load Balancer (ELB) / Nginx / Cloudflare
- **Monitoring & Logging:** Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana)

2.2 Deployment Environments

Environment	Purpose	Hosting Platform
Development	Ongoing Development	Local or Cloud Dev Instance
Staging	Pre-production testing	Cloud Staging Instance
Production	Live System for end users	Cloud Production Instance

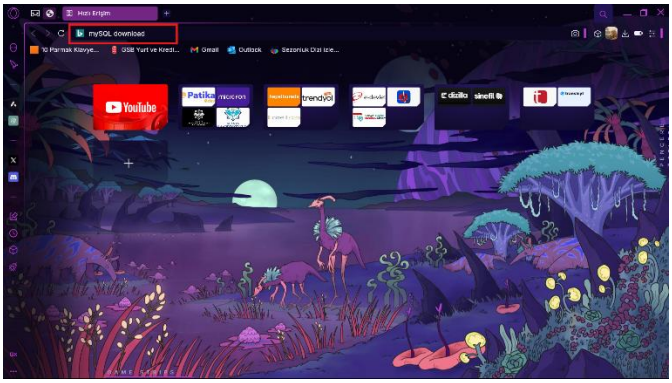
3. Deployment Process

3.1 Continuous Integration & Deployment (CI/CD)

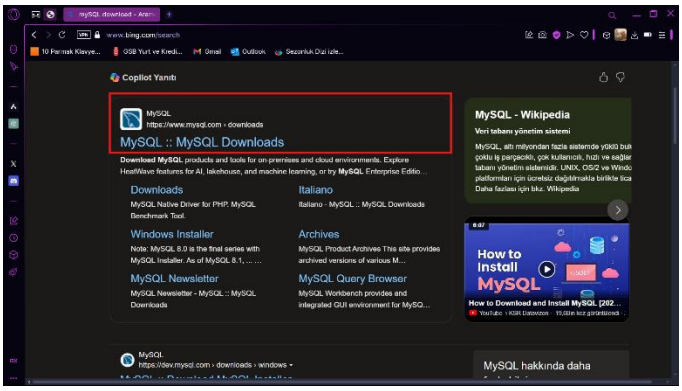
Tools Used: MySQL Workbench Visual Database Designer

Process Steps:

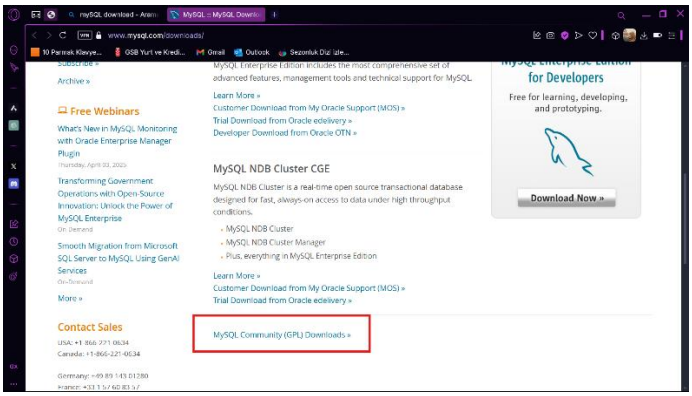
1. Type “MySQL download” into any browser.



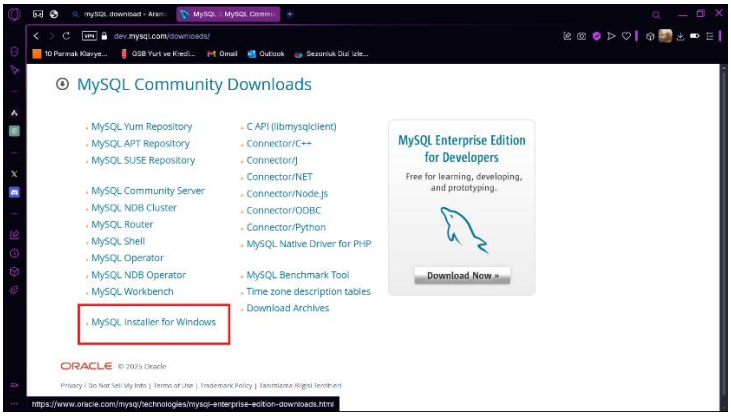
2. Click on the site shown in the window.



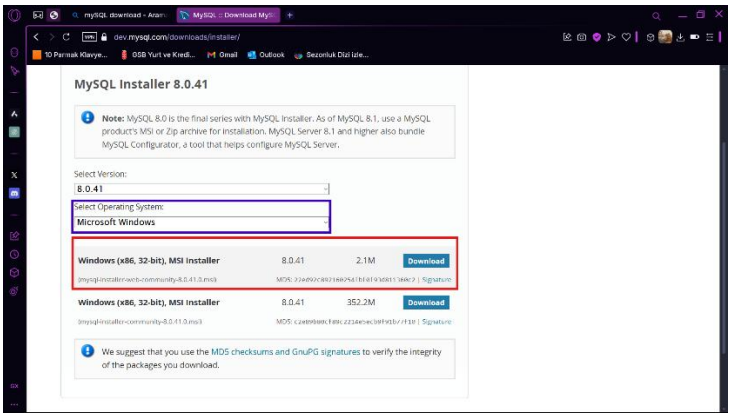
3. Scroll down the page and click on the section indicated.



4. Click on the part labeled “MySQL Installer for Windows”.



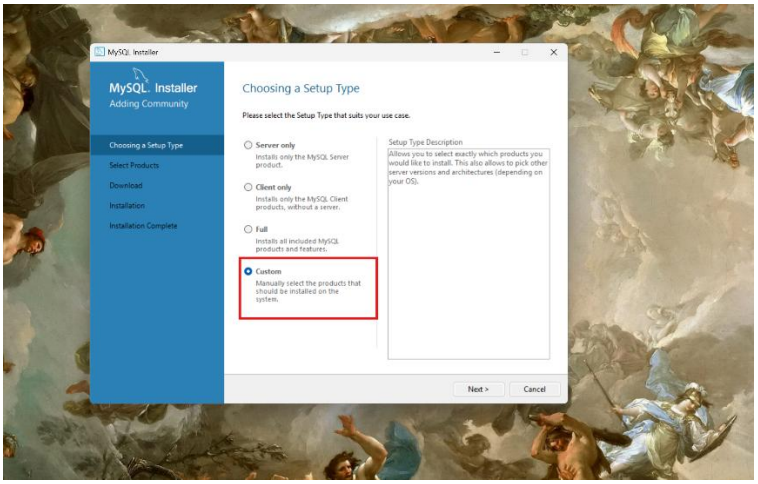
5. In the blue box, select the operating system and click the “download”.



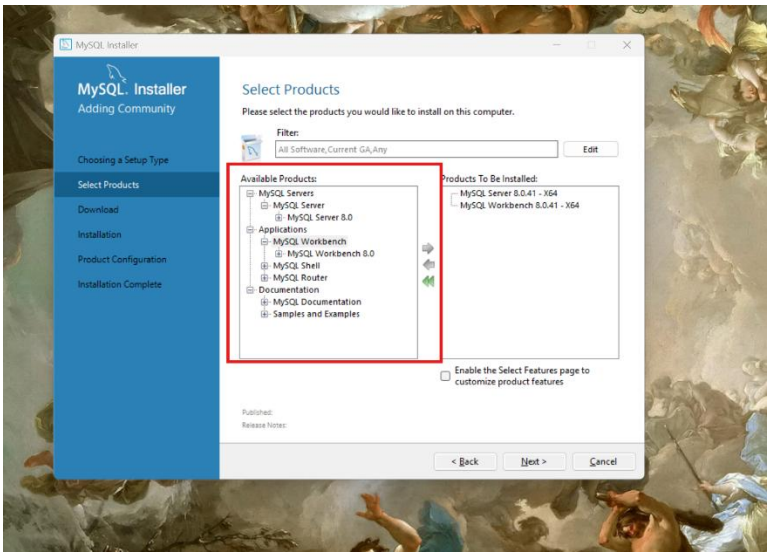
6. Once the download is complete, open the application and start the installation.



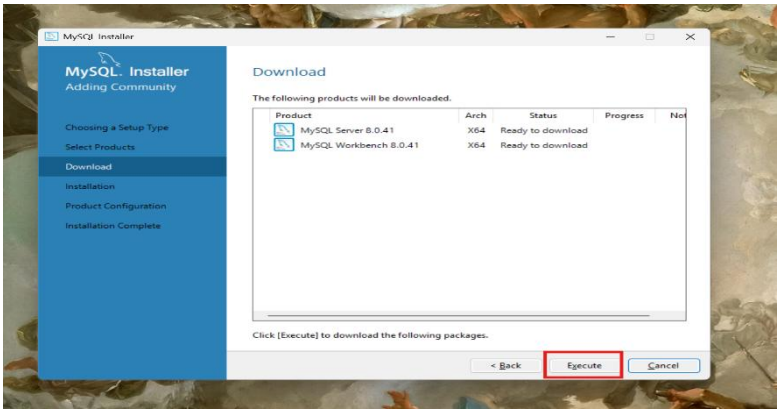
7. In the opened window, select “custom” and click “next”.



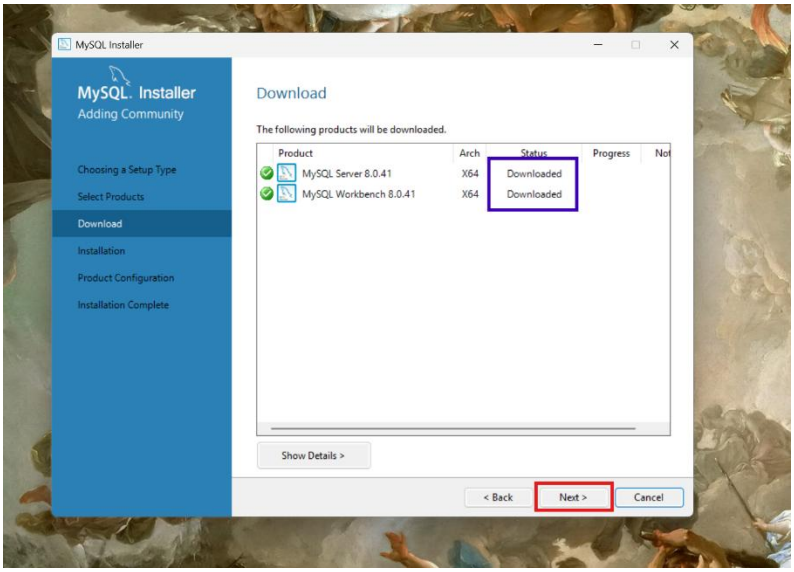
8. Select the downloaded version from the “MySQL Servers” section, and select “MySQL Workbench” from the “Applications” section. Then, click the topmost arrow in the middle and proceed to the next step.



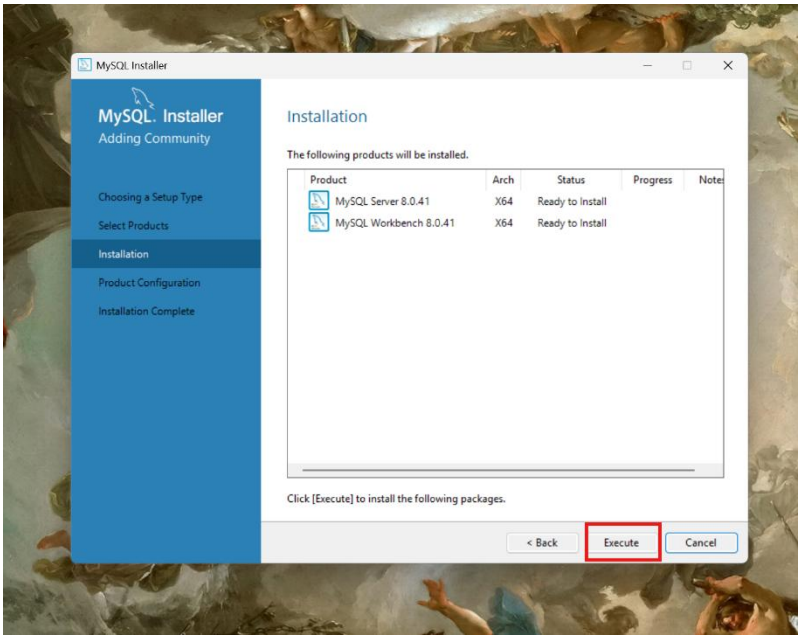
9. Click on the “Execute” and wait for the downloads to complete.



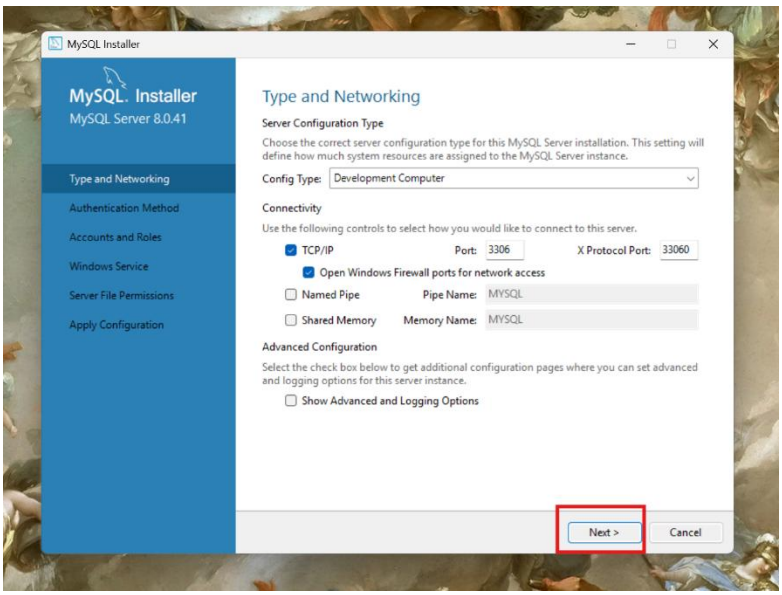
10. After the downloads are completed, proceed to the next step.



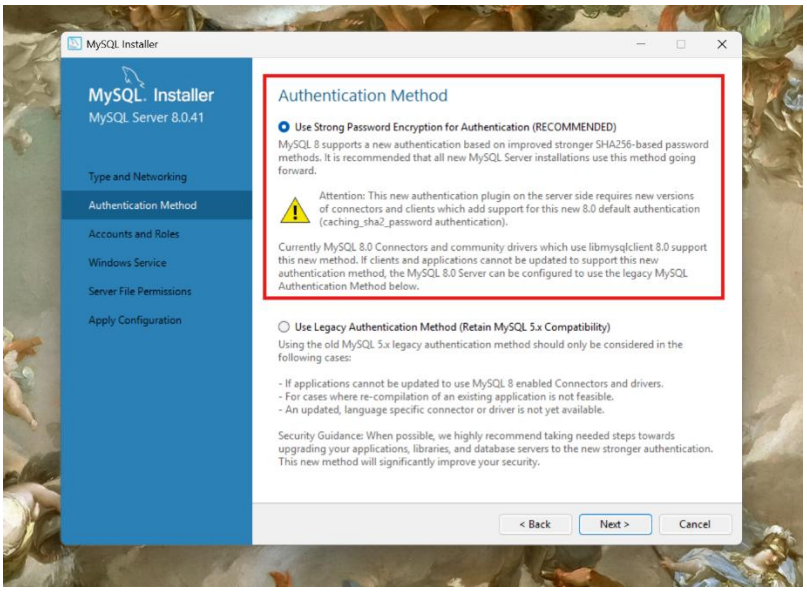
11. Click on the “Execute” to start the installation.



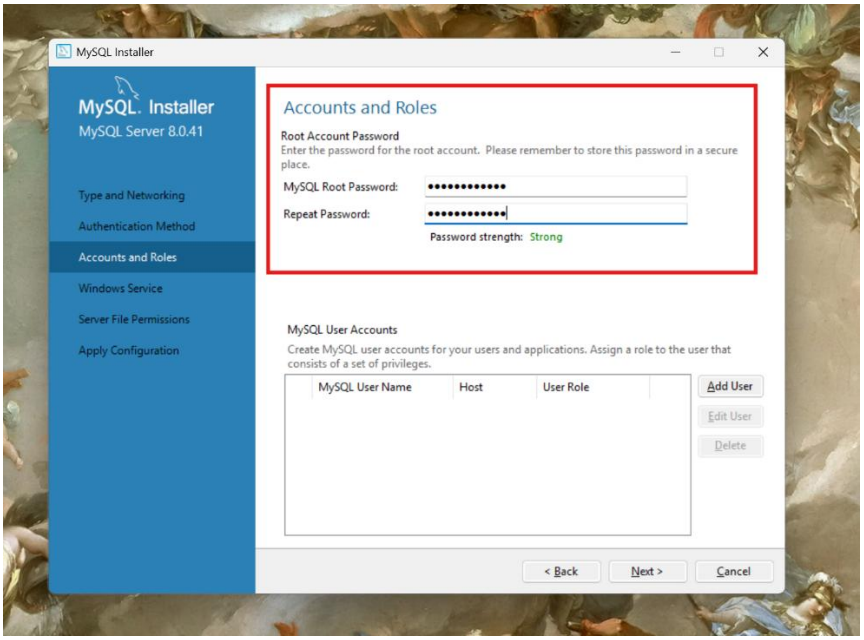
12. Move to the next step without changing anything in this window.



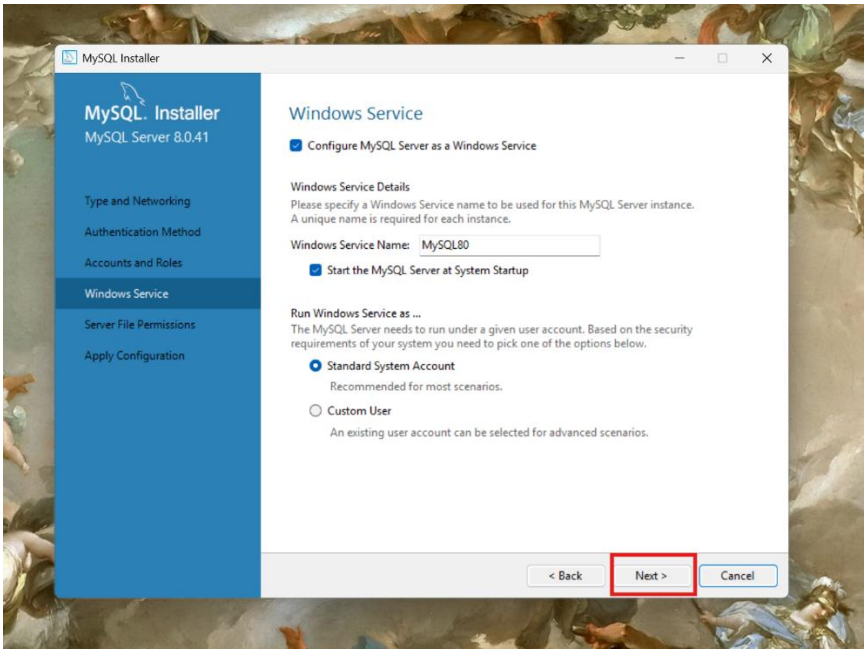
13. Continue with the recommended settings to set a password.



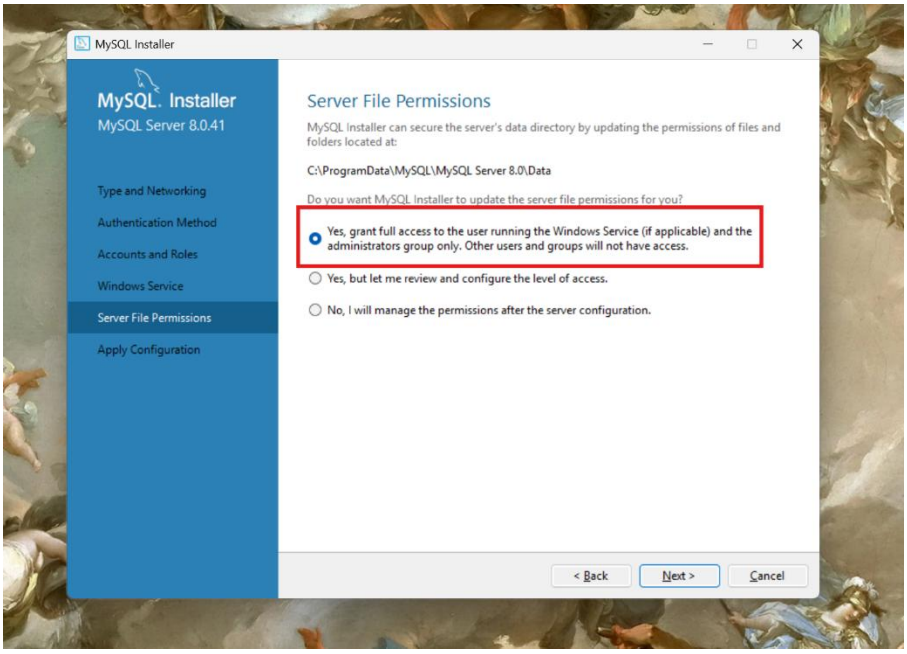
14. Set a password and proceed to the next step.



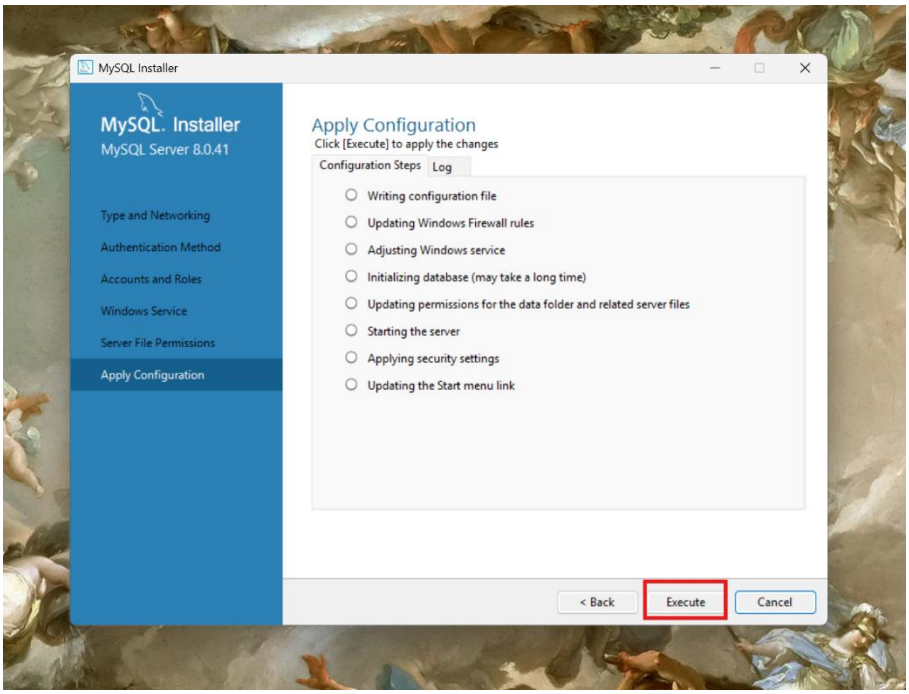
15. Move to the next step without changing anything in this window.



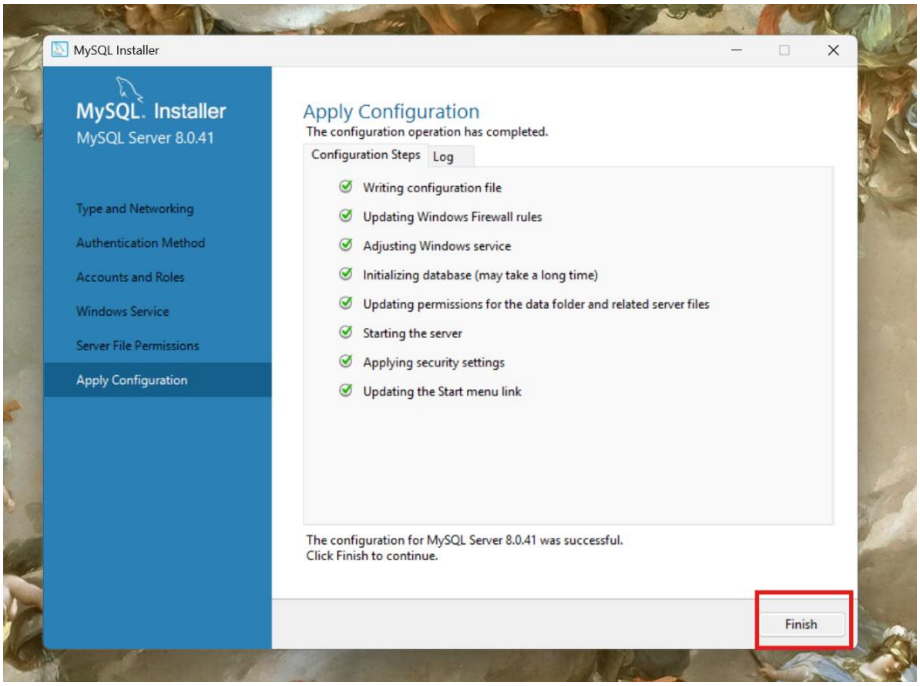
16. Grant the necessary permissions and proceed to the next step.



17. Click the “Execute” and wait for the processes to finish.



18. Once the processes are finished, complete the installation by clicking the “Finish”.



3.2 Deployment Strategies

Strategy	Description
Blue-Green	Traffic is switched to a new environment post-verification.
Rolling Update	Gradually deployment with zero downtime.
Canary Release	Limited initial rollout before full deployment.

3.3 Rollback Strategy

- **Database Backups:** Automatic backups are taken before deployment.
- **Feature Flags:** Features can be toggled off in case of issues.
- **Version Control:** Rollback to the last stable deployment.

4. Monitoring & Logging

4.1 Monitoring Tools

- **Application Performance Monitoring:** New Relic / Datadog
- **Infrastructure Monitoring:** Prometheus / Grafana
- **Error Tracking:** Sentry / ELK Stack

4.2 Logging Framework

- **Centralized Logging:** Logstash & Kibana are used for log aggregation.
- **Retention Policy:** Logs are retained for 90 days.

5. Security Considerations

5.1 Authentication & Authorization

- JWT-based authentication for secure user login.
- Role-Based Access Control (RBAC) for module-level permissions.

5.2 Data Protection

- Encryption in transit and at rest (AES-256, TLS 1.2+).
- Regular vulnerability scanning and security audits.

5.3 DDoS Protection

- Mitigation through Cloudflare / AWS Shield.
- API rate limiting to prevent abuse**.**

6. Backup & Disaster Recovery Plan

6.1 Backup Strategy

- Daily full database backups, hourly incremental backups.
- Version-controlled backups for static files.

6.2 Disaster Recovery Plan

- **Failover Strategy:** Multi-region setup for high availability.
- **Recovery Time Objective (RTO):** Less than 15 minutes.
- **Recovery Point Objective (RPO):** Less than 5 minutes.

7. Conclusion

This deployment plan ensures a reliable, secure, and scalable setup for the Hospital Management System, with zero downtime, robust monitoring, and a well-defined rollback process. The plan is tailored to integrate the key modules—Doctor, Patient, Prescription, Appointment, and Polyclinic—into an efficient, user-friendly system.